J Child Orthop (2007) 1:313–322
DOI 10.1007/s11832-007-0057-x

ORIGINAL CLINICAL ARTICLE

Characteristics of patients with congenital clasped thumb: a prospective study of 40 patients with the results of treatment

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Received: 12 June 2007 / Accepted: 20 September 2007 / Published online: 17 October 2007
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Abstract
Purpose Congenital clasped thumb is a deformity that is associated with heterogeneous congenital anomalies and it has been addressed in many congenital syndromes. The aim of this study was to diagnose and evaluate cases of clasped thumb as regards the associated congenital anomalies and syndromes, and evaluation of the results of treatment of such cases.
Methods A prospective study on 40 patients with 73 clasped thumbs was done. All the patients' data regarding their personal, family, pregnancy and developmental histories were recorded. All the patients were exposed to thorough clinical and radiological examination and genetic assessment. The cases were classified using the Tsuyuguchi et al. (J Hand Surg [Am] 10:613–618, 1985) classification into three types. Conservative treatment was adopted in ten hands, and surgical treatment was performed for 28 hands in 17 patients, with an average follow-up of 26 months.
Results Positive consanguinity was recorded in 57.5% of cases. Associated anomalies were recorded in 77.5% of cases. Type I was the most common one, followed by type III and then type II. Conservative treatment is effective in type I cases when presented early, and all patients were satisfied with the results of surgical treatment.
Conclusions We reported associated anomalies which are to our knowledge have not mentioned before in the literature which include; congenital blindness, radial deviation of the index finger and ventricular septal defect. We found that 68% of the patients had associated syndromes, and this has not been mentioned before. In this study, we found that there were no difference between type II and type III clasped thumb as regards the pathological findings, severity, the operative procedures, the treatment protocol and the operative results. Properly planned treatment gives satisfactory results.

Keywords Thumb · Clasped thumb · Thumb contracture · Flexed adducted thumb

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Introduction

A persistent flexed adducted thumb after the 3rd or 4th month of life is called congenital clasped thumb [1]. It is due to deficiency of the extensor pollicis brevis or longus or both. The deformity is usually accompanied with a variable degree of narrowing and contracture of the first web space [2]. This deformity is usually associated with other generalized musculo-skeletal malformations, commonly; arthrogryposis, digitotalar dysmorphism and Freeman–Sheldon syndrome.

Most of the literature has discussed a surgical technique or case reports of associated anomalies or syndromes. Little has been written about the epidemiology of the deformity.

The aim of this study is to characterize a group of patients describing the associated anomalies and syndromes, the possible predisposing factors and results of treatment.

Patients and methods

Forty patients with 73 congenital clasped thumbs were prospectively studied. There were 29 males and 11 females, with an average age at presentation of 27 months (range from 3 months to 11.5 years). The condition was bilateral in 33 cases.

The patients were exposed to thorough assessment, including recording of the family history, pregnancy history and full clinical assessment performed by a paediatrician. Genetic assessment was intended to be done for all cases by a paediatric geneticist (last author). Radiographic examinations of the extremities, skull, spine, and the pelvis and abdominal ultrasonography were conducted for all the patients. Echocardiography was done when recommended by the paediatrician.

The deformed thumbs were classified according to Tsuyuguchi et al. [3] into three types (Fig. 1).

Thirty-eight hands were treated. Non-operative treatment was followed in ten hands in five patients who presented before the age of 1 year (eight hands type I and two hands type II). Surgical treatment was done in 28 hands in 17 patients (cases presented later than 1 year of age or which had not responded to non-operative treatment) (Tables 1, 2).

Table 1 The distribution of the operated patients with regard to sex and bilaterality

	No. of patients	Bilateral operations	Unilateral operations
Male	13	8	5
Female	4	3	1
Total	17	11	6

The cases which were not treated were; nine hands waiting for surgery, 16 hands where the patients did not return for surgery, and ten hands where the parents refused surgery.

Treatment protocol

- Non-operative treatment: full time splinting of the thumb in extension for at least 6 months, followed by night splinting for further 6 months after achieving active extension of the thumb.
- Operative treatment: surgical procedures for every case were variable according to the degree of narrowing of the first web, stability of the metacarpophalangeal (MP) joint and muscle deficiency.

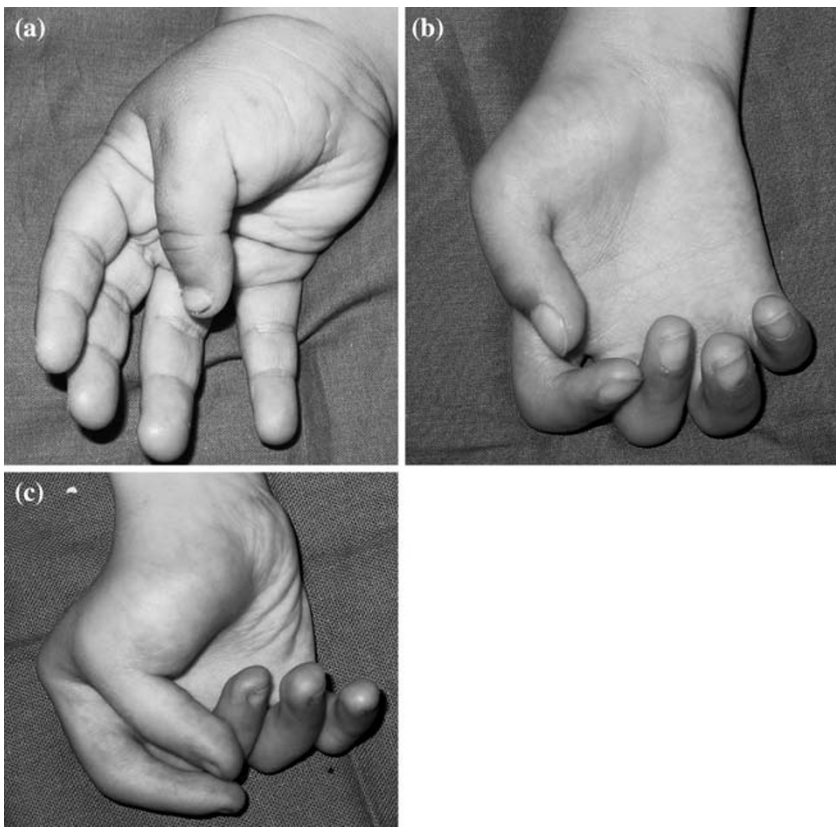


Fig. 1 **a** Type I: supple clasped thumb. The thumb could be passively abducted and extended against the resistance of thumb flexors, without other digital anomalies. **b** Type II: clasped thumb with hand contractures. The thumb could not be passively extended and abducted, with or without other digital anomalies. **c** Type III: clasped thumb associated with arthrogryposis [1]

Table 2 The number of different types of clasped thumb operated upon

Type of clasped thumb	No. of operated hands
I	8
II	8
III	12

Surgery aims at widening the narrow first web space, including the skin and deep tight structures. Skin widening was done through either simple Z-plasty, four-flap Z-plasty [4], butterfly flap [5], dorsal rotational advancement flap [6] and modified dorsal rotational advancement flap [7]. The choice of the technique is variable according to the degree of narrowing. Through the skin incision designed to widen the web, the dissection is deepened through underlying fascia over the intrinsic muscles, protecting the distal arborisation of the superficial radial nerve, the flexor tendon and neurovascular bundles to the index finger. The tight structures are identified to be released; the origin of adductor pollicis muscle is released from the third metacarpus. If necessary, the first dorsal interosseous muscle is elevated from the first metacarpus. The thumb is then manipulated into extension and abduction. If necessary, the carpometacarpal joint capsule is released. After achieving full release, the first metacarpus is maintained in full abduction with two crossed k-wires across the first web space.

A markedly unstable MP joint necessitated chondrodesis, but in cases of mild laxity of the ulnar collateral ligament, double breasting of the capsule was done. In either case, the MP joint was transfixed with a single or two crossed k-wires.

Tendon transfer was done to restore active extension of the MP joint when it was stable or after achieving stability with reconstruction of the ulnar collateral ligament. The extensor indicis was the preferred tendon for transfer, and if it was absent, one slip of the flexor digitorum superficialis muscle is transferred to the vestigial remnant of the deficient thumb extensors (Table 3).

An above-elbow splint was applied immediately. The k-wires were removed after 6 weeks of surgery. The position was maintained in a night splint with the thumb extended for at least 6 months post-operatively, with daytime active use of the thumb being encouraged. The average duration of follow-up was 26 months (range from 16 to 60 months).

The criteria used to assess results of treatment were:

I: Parents' satisfaction: regarding cosmetic appearance and function.

II: Thumb position and appearance: degree of abduction and rotation.

Table 3 The different surgical procedures done

Procedure	Type of operation	No. of hands
Widening of the first web space	Z-plasty	5
	Four-flap Z-plasty	4
	Butterfly flap	1
	Dorsal rotational advancement flap	6
	Modified dorsal rotational advancement flap	10
Release of first web structures	Release of fascia of first web space	24
	Release of the adductor pollicis muscle	13
	Release of first dorsal interosseous muscle	9
	Release of the carpometacarpal joint capsule	9
Chondrodesis of the MP joint		18
Ligament reconstruction of the MP joint	Double breasting of the ulnar capsule and ulnar collateral ligament	2
Z-lengthening of the flexor pollicis longus tendon		2
Tendon transfer	Extensor indicis transfer	7
	Flexor digitorum superficialis of the middle finger	1

III: Stability of MP joint.
IV: Thumb function: degree of opposition and the ability to grasp different objects.

The degree of abduction, rotation, stability and opposition were graded into four grades according to Gilbert (personal communication) (Table 4).

Results

The results will be discussed under the following headings:

I: Clinical findings
II: Operative findings
III: Results of treatment

Clinical findings

Family history

Twenty-three patients had positive consanguinity. First-degree consanguinity was found in 13 patients and far relationship in ten patients. Parents' anomalies were found in six patients; the most common parents' anomalies were club feet (two patients) and windblown deformity of the hands (two patients).

Seven patients had either the brother or the sister affected with other congenital anomalies. One patient had a sister with clasped thumb, and another one had three brothers who had other congenital anomalies but they could not be assessed. One patient had the mother, the aunt and the grandfather with windblown deformity, and his cousin had clasped thumb.

Prenatal history

Positive history of fever during pregnancy was reported in seven patients, but the mothers could not specify the type of infections they had had. One patient had a history of radiation exposure during pregnancy. Drug intake during the first trimester was found in 12 patients. Most of the mothers could not specify the types of medication they received (Table 5).

Table 5 Types of drugs taken during the first trimester of pregnancy in 12 patients

Type of drug	No. of patients	Reason for intake
Contraceptive pills	1	Continue intake (unknown pregnancy)
Antihistaminic	2	Common cold
Antibiotics	1	Fever
Analgesics	2	Pain relief
Antacids	1	Gastritis (unknown pregnancy)
Unknown	5	

General examination

The distribution of different types of clasped thumb and their sex distribution are mentioned in Table 6. Thirty-one patients (58 hands) were found to have associated anomalies.

Facial abnormalities were found in 18 patients; ptosis, antimongoloid obliquity of the palpebral fissure, cleft palate, broad bossed forehead, flat face, depressed nasal bridge and long philtrum. Whistling face was diagnosed in six patients. Two patients had congenital blindness. Four patients had short neck, and one had additional webbing. Congenital heart anomalies were found in two patients; one had atrial septal defect and the other had ventricular septal defect. Scoliosis was found in one patient with arthrogryposis, but MRI was not done. No abdominal anomalies were found in any patient, but accidentally discovered renal stones were found in two patients (Tables 7, 8, 9).

Associated syndromes

In 11 patients, genetic assessment was refused by the parents, and their associated syndromes were not diagnosed; two of them had congenital blindness with clasped thumb. Different syndromes were present in 20 patients out of the 29 patients assessed (68%) (Table 10).

Table 4 Gilbert's method of assessment of thumb function (personal communication)

Abduction	Rotation	Stability	Opposition	Results
40°–45°	110°–120°	Very stable	With little	Excellent
30°–40°	90°–100°	Stable (stable at the ulnar side)	With ring	Good
10°–30°	80°–90°	Mild instability (no problem at pinch)	With middle	Fair
0°–10°	<80°	Unstable	None	Poor

Table 6 The different types of clasped thumb with sex distribution

Type	Male	Female	Total
I	22	8	30
II	5	9	14
III	25	4	29

Table 7 Lower limb anomalies associated with clasped thumb

Associated lower limb anomaly	No. of patients	Type of clasped thumb
Bilateral hip dislocation	3	III
Unilateral hip dislocation	4	III
Congenital knee dislocation (bilateral)	2	III
Stiff knee in extension	3	II, III
Congenital vertical talus (bilateral)	11	I, III
Congenital vertical talus (unilateral)	1	II
Congenital talipes equinovarus (bilateral)	7	I, III

Table 8 Upper limb anomalies associated with clasped thumb

Associated upper limb anomalies	No. of patients	Type of clasped thumb
Limited shoulder movements	5	III
Pterygium cubitale	1	III
Limited elbow movements	2	III

Table 9 Associated hand anomalies

Type of associated anomaly	No. of patients
Bilateral ulnar drift hand	Four patients (10%) (8 hands)
Unilateral ulnar drift hand (Fig. 2)	One patient
Flexion deformity of the four fingers (Figs. 1b, 3)	Twelve patients (30%) (24 hands)
Flexion deformity in less than four fingers	eight patients (20%) (11 hands)
Extension deformity of the wrist (Fig. 1c)	two patients (5%) (4 hands)
Radial deviation of the index finger (Fig. 3)	One patient in left side (2.5%)

Table 10 Associated syndromes

Syndrome	No. of patients	Type of clasped thumb
Freeman–Sheldon syndrome	6	III
Arthrogryposis multiplex congenita	5	III
Multiple pterygium	3	III
Digitotalar dysmorphism	2	I
Clasped thumb club foot	2	I
Congenital contractural arachnodactyly	1	II
Emery–Nelson syndrome	1	II

Patterns of inheritance among genetically assessed cases

Autosomal dominant inheritance: there was variable expressivity of the autosomal dominant inheritance in one parent and sibs of 13 patients. These patients had isolated Freeman–Sheldon syndrome (four patients), clasped thumb deformity (three patients), arthrogryposis multiplex congenita (two patients), clasped thumb club foot syndrome (two patients), congenital contracture arachnodactyly (one patient) and digitotalar dysmorphism (one patient).

Suspected autosomal recessive inheritance: 13 patients had positive family history of consanguinity without a family history of similar conditions. These patients have isolated clasped thumb deformity (six patients), arthrogryposis multiplex congenita (three patients), multiple pterygium syndrome (three patients), and Emery–Nelson syndrome (one patient).

Sporadic hands: three patients had no consanguinity or similar familial anomaly; two of them have Freeman–Sheldon syndrome, and the other have digitotalar dysmorphism.

Operative findings

Abnormal pathological anatomy

Attenuated extensor pollicis longus and brevis were found in all hands, and they appeared as vestigial strands of tendon-like structures measuring about a millimeter in diameter. Absent extensor indicis was detected in one out of eight hands with tendon transfer.

Contracture of adductor pollicis muscle was found in 13 hands (seven hands had Type III and six hands had Type II clasped thumb). Meanwhile, first dorsal interosseous muscle was contracted in nine hands (six hands had Type III and three hands had Type II clasped thumb).

Global instability of the MP joint was present in 18 hands. Laxity of ulnar collateral ligament of the MP joint was detected in two hands (Type III clasped thumb). The true nature of instability was identified after release of the web space.

Abnormal articular surface of the first MP joint was detected in six hands (four hands had Type II and two hands had Type III clasped thumb) with hypoplasia of the volar aspect of the first metacarpal head. The dorsal capsule of the MP joint was adherent to the cartilage of the metacarpal head and sharp dissection was needed to separate it in eight hands (four hands Type II and four hands Type III clasped thumb). Flexion contracture of the MP joint was present in eight hands (four hands Type II and four hands Type III).

Post operative complications

Superficial necrosis of the tip of the modified dorsal rotational advancement flap with venous congestion occurred in one hand Type II, and it was treated with repeated dressings without effect on the result. Proximal retraction of the tip of the dorsal rotational advancement flap was noticed in all cases. This produced some re-narrowing of the web, which was avoided in later cases by using the modification of this flap [5]. Non-union of chondrodesis of MP joint occurred in two hands (Figs. 2, 3).

Results of treatment

Results of operative treatment

Twenty-eight hands were assessed postoperatively with regard to:

(1) Parent's satisfaction. The parents of all the patients were satisfied with the results (Fig. 4).

(2) Cosmetic appearance. The simple Z-plasty deepened the web, transforming it into a slit, and did not give the normal rounded appearance of the web. The appearance of the first web space was better with the other techniques. The modified dorsal rotational advancement flap allowed maximal degree of widening more than the other techniques used. Twenty-four thumbs had excellent abduction, and four thumbs had good abduction. Twenty-six thumbs had excellent rotation, and two had good rotation.

(3) Stability of the MP joint. The stability of the MP joint of the thumbs operated upon was excellent in 24 cases. In the two hands with non-union of the chondrodesis, revision of chondrodesis was needed in one thumb with poor stability, while in the other the stability was fair and there was no need for surgery; otherwise, the rest of cases with chondrodesis had excellent stability (Fig. 4). The stability of two cases with ligamentous reconstruction of the ulnar collateral ligament was graded as good.

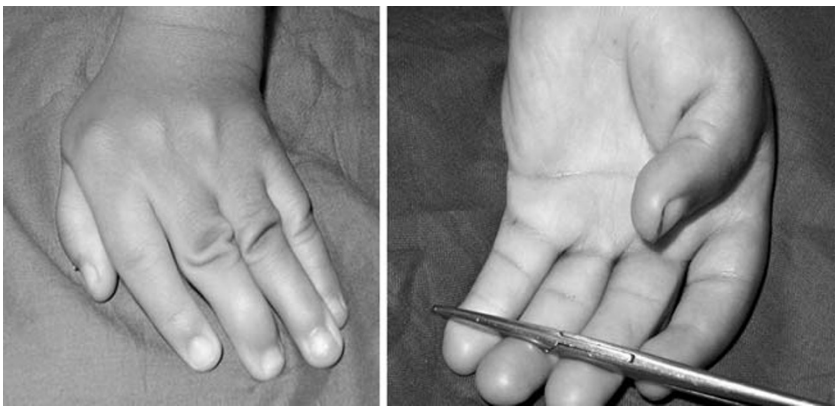


Fig. 2 Ulnar drift hand

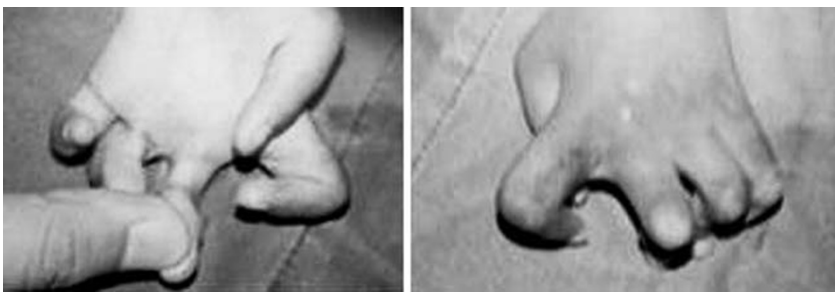


Fig. 3 Radial deviation of the index finger associated with flexion deformity of the lateral four fingers



Fig. 4 **a** Severe type III clasped thumb in patient with arthrogryposis treated surgically with release of the first web, widening the web with modified dorsal rotational advancement flap and chondrodesis of the MP joint of the thumb. **b** Result of surgical treatment

(4) Thumb functions. Active extension of the thumb was full in all cases for which tendon transfer was done (Fig. 5).

Ten thumbs had excellent opposition, 14 thumbs had good opposition and two thumbs had fair opposition. Two thumbs were unable to oppose, and opponensplasty was needed. In all the hands for which muscle transfer to the extensors of the thumb was done, the opposition were excellent. There was improvement of the grasp pattern in all the operated thumbs. Twenty-five hands were able to pick up a pen and a key and were able to grasp a ball. Two hands (one patient) wer able to pick up a key and a pen between the thumb and the middle finger because of the severe flexion deformity of the index finger. In one hand, the ability to grasp and pick up an object could not be assessed.

Results of non-operative treatment

Non-operative treatment was successful in eight hands type I, with achievement of full active extension of all joints of the thumb. There was no improvement in the two hands of type II that will need surgical treatment.

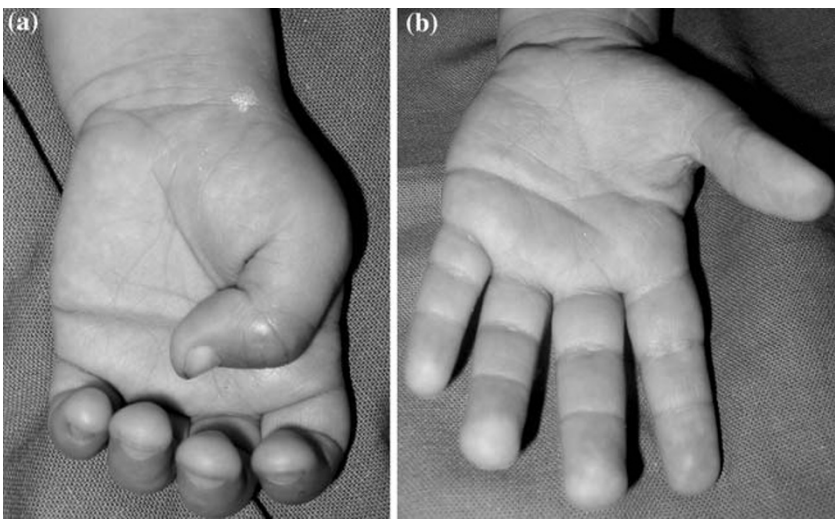


Fig. 5 **a** Type I clasped thumb. **b** Result of surgical treatment after transfer of extensor indicis to remnant of the extensor tendons

Discussion

Congenital clasped thumb is a deformity that is associated with heterogeneous congenital anomalies and syndromes [1]. The incidence of these anomalies and syndromes vary, and are based on study of small groups of patients. Little has been written about the distribution and epidemiology of clasped thumb. Also, the characteristics of the patients with clasped thumb are not well described in the literature.

The clinical evaluation of the patients in this study revealed that the male-to-female ratio is 2.5:1, which is similar to the reported ratio by Lin et al. [8], and is near to the estimated ratio reported by Flatt [9]. However, it differs from the 1:1 ratio reported by Tsuyuguchi et al. [3]. This difference can be attributed to the wide variety of syndromes associated with clasped thumb and their different modes of inheritance, and also the small number of reports in the literature that document the sex ratio of the affected cases.

With regard to the possible predisposing factors, the incidence of positive consanguinity is 60% and positive family history in our study is 32.5%, which is much higher than the 10% positive family history that was reported by Tsuyuguchi et al. [3], who also did not mention the incidence of consanguinity.

The high percentage of positive consanguinity and family history in about one-third of our cases, and the high incidence of bilateral cases, strongly suggest that the genetic defect is an important causative factor [10]. However, there may be an underestimation of the percentage of drug intake. We noticed that mothers might deny intake of drugs taken to abort babies. Also, the drug may be in the form of a herbal extract, which may be taken for the same purpose. In the majority of cases, other therapeutics were taken without knowing the existence of pregnancy.

One-third of cases were sporadic. This agrees with the literature regarding the causative factors for such an anomaly [3]. However, the exact causative factors are not yet known.

This study confirms the previously reported high incidence of associated congenital anomalies with clasped thumb [1, 3, 11–25]. About 77.5% of our patients were found to have associated anomalies; this value differs according to the number of syndromes included in the literature, and no previous literature studied the incidence of congenital anomalies with clasped thumb, as the different syndromes were studied separately or the literature contains case reports of certain syndromes. The incidence of upper limb anomalies in our patients (15%) agrees with reported cases of Temtamy and McKusick [1].

Other anomalies encountered included congenital heart disease in the form of ventricular septal defect—which was detected in one patient who had no other anomalies except unilateral type I clasped thumb—and this association was not found to be reported in the literature. We reported other anomalies, which to our knowledge have not been mentioned before which include two patients with congenital blindness in association with bilateral clasped thumb, and a case with radial deviation and rotation of the index finger. No abdominal anomalies were encountered in this study, and this agrees with the literature [1, 17, 26].

Associated syndromes were encountered in 68% of patients, and this percentage was not studied before in the literature.

We used the classification suggested by Tsuyuguchi et al. [3]. This classification was also used by Mih [27]. We found that there was no difference between type II and type III clasped thumb with regard to the pathological findings, severity, the operative procedures, the treatment protocol and the operative results; thus, it was wiser to classify the clasped thumb into supple and complex types, as McCarroll [28] did.

The distribution of different types of clasped thumb and the male to female distribution was almost the same as reported by Tsuyuguchi et al. [3] (Table 11).

Our protocol of management is similar to that followed by Weckesser et al. [10], Lipskeir and Weizenbluth [29] and McCarroll [28], who divided the clasped thumb into two types (supple and complex), and treated the supple type with splinting and tendon transfer if splinting failed, while the complex type was treated by correction of the fixed contractures and reconstruction of lax ligaments and tight skin as appropriate. Tsuyuguchi et al. [3] cases were treated by splinting for types I and II cases and operative treatment for type III and type II cases which didn't respond to conservative management.

In this study, 50% of operated cases were type III clasped thumb; 29% of cases were type II, and only 21% were type I cases who were presented at an age older than 1 year. This differs from the cases presented by Tsuyuguchi et al. [3], where 50% of the patients were type I and 50% were type II; this can be explained by the young age at presentation of their type I cases, as all their patients were within their first 7 months of life.

The operative findings agreed with Crawford et al. [2] and McCarroll [28], in that the extensor tendons were not

Table 11 Comparison between the distributions of different types of clasped thumb in the present study and the study presented by Tsuyuguchi et al. [3]

Type	Present study (%)	Study of Tsuyuguchi et al. (%)
I	38	32
II	19	28
III	43	40

absent but attenuated in the three types. Our findings also agreed with McCarroll [28], in that the complex type of clasped thumb (types II and III) have contractures in the soft tissues in the volar surface of the digit, involving all tissues including skin, subcutaneous fascia, periarticular structures and in the radial collateral ligament–volar plate complex of the MP joint. The dorsal capsule of the MP joint is adherent to the cartilage of the metacarpal head, and there is abnormality of the articular surface of the metacarpal head in some cases of types II and III clasped thumb. The shortening of the flexor pollicis longus muscle was present in cases of types II and III, but Z-lengthening was needed in few cases because in the majority of cases we did chondrodesis of the MP joint.

We noted that the severity of these pathological findings is variable, and dependent on the age of the patient at the time of surgery. We also noted that these findings were present in types II and III clasped thumbs, and there was no difference in these pathological findings in both types.

We used different skin incisions in this study to widen the web space. No technique can be standardized for release of the web as this depends on the severity of web narrowing and degree of contracture; as a result, it is difficult to compare the different methods used for release.

There were no universal criteria for the evaluation of the results of management of clasped thumb, due to the difficulty in assessing the thumb function at that young age, and other authors used different systems for evaluation of their results. Some authors used the degree of active extension of first MP joint as the reference for evaluation [8, 10]. Tsuyuguchi et al. [3] added the degree of active radial abduction of the trapeziometacarpal joint to their system of evaluation. Lipskeir and Weizenbluth [29] added the width of the first web space to their scoring system, and they mentioned that active extension of first MP joint is the most important factor for the apprehension of large objects.

We used a combination of criteria to evaluate the thumb function, because it is of no benefit to achieve active thumb extension without achieving stability of the MP joint or without widening of the web. It is very important to consider the stability of the MP joint. Using the active extension as the sole criteria for assessment is possible in type I cases where this is the only deficient function.

The results of splinting in this study were excellent in type I clasped thumb [3, 8]. The small number of patients that followed the non-operative treatment in this study is attributed to the delayed presentation of the patients until after the age of 1 year.

In this study, the results of tendon transfers done for type I clasped thumb were excellent. We did not use an osseous tunnel in the proximal phalanx for the extensor indicis tendon as originally described [4], and instead we sutured it to the attenuated extensor tendon of extensor pollicis brevis, as described by Mih [27].

In the case of ulnar collateral ligament instability of the MP joint, ligamentous stabilization is a prerequisite for tendon transfer. Although the ligament reconstruction did not give excellent stability, the residual instability did not interfere with thumb function. In the case of global instability of the MP joint, chondrodesis is the best way to achieve stability, and at the same time it obviates the need for tendon transfer. The results of chondrodesis are better than that reported by Tsuyuguchi et al. [3] and Lipskeir and Weizenbluth [29].

In this study, we found that properly planned treatment according to the type of the deformity improves the cosmetic appearance and functional capabilities of the hand.

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